DATA/CS 172: Python for Data Analysis

Matplotlib Part III: Multiple plots and more

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10/28/2024

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Announcements

- Keep on top of labs
- Start on Program 2

Today

- A few more tips with single plots
- Figures and subplots

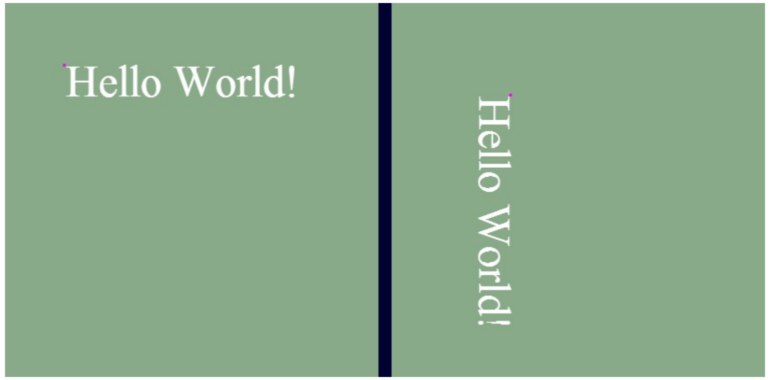
Last time, we saw how to make bar charts…

- They looked pretty good (especially if we have a single type of bar)
- But what if we have a lot of categories on the x-axis?

X rotation

- Best way to fix this is by rotating the x labels
- plt.xticks(…, rotation=degrees)

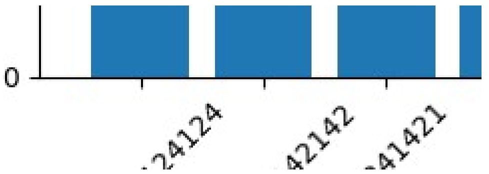
- If you have a TON of labels, degrees=90 may be your best bet
  - If just a slight amount, should not go that extreme (takes up a lot of space)



Problem #1

- With a lot of long labels, and degrees=90…
    - The graph is cut off!
- How can we see it?
    - Need to add plt.tight_layout() to resize view box
        - Before showing/saving

Problem #2



- If we choose a lower angle- hard to match up label to bar
- Solution: Change horizonal alignment (ha keyword arg to xticks)
- Default is ‘center’, but should chose left or right
  - Right may make more sense

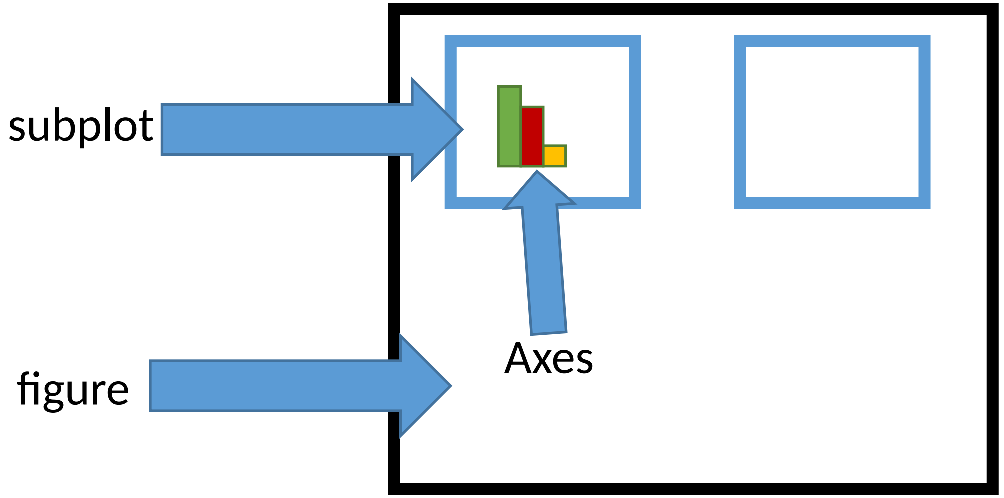
Another issue…

- So far, we have called plt.show() after every graph
- Or done plt.savefig on a single graph
- What if we call plt.savefig on multiple graphs?
  - E.g. script to generate result graphs for paper

- Leaves the old graph on top of the new one!
  - Need to use plt.cla() & plt.clf() to clear graph

Multiple plots

- Often it can be useful to present multiple graphs
  - Side-by-side
  - In a grid layout etc.
- To do that in matplotlib, the outer thing is called a figure
- And the inner thing is called a subplot



How do we set this up?

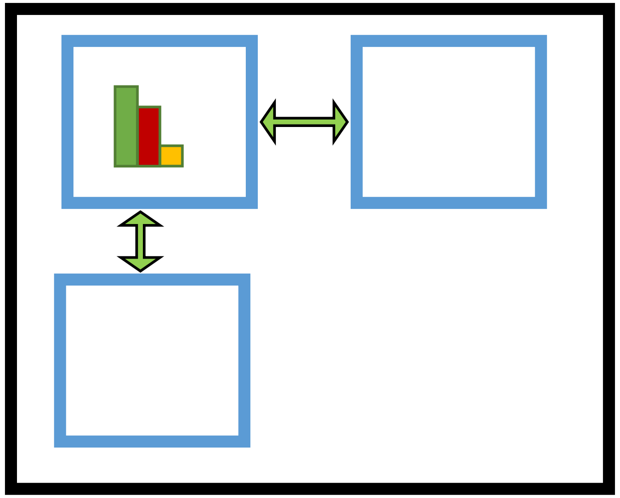
- Step 1: Decide what our grid is going to look like
  - numRows
  - numColumns
- Step 2: create a new figure with those subplots
  - (fig, axes) = plt.subplots(numRows, numColumns)
- Step 3: get the axis of interest
  - axes[rowIndex, colIndex]

Drawing on subplots

- The subplot commands returns axes
  - Commands should be run on the appropriate axes object instead of plt:
  - Ax1 = axes[rowIndex, colIndex]
- Can then run most plt.* commands on the Axes
  - ax1.plot(…)
  - ax1.bar(…)
  - Some of them need to be prefixed with set_
    - set_title()
    - set_xticks()
      - Works like plt.xticks()
    - set_xticklabels() [Highly Discouraged]

Adjusting spacing

- Default spacing is pretty generous
- Can be a problem if you're trying to cut down on space (for a paper etc.)
- fig.subplots_adjust(wspace=0.1, hspace=0.1)
- shareaxes = true



Changing size of a figure

- plt.subplots(numRows, numCols, figsize = (xInches,yInches))
- Matplotlib assumes dpi of monitor
  - If we know dpi can do better:
    - plt.subplots(numRows, numCols, figsize = (10,8), dpi =96)

Recap

- Matplotlib can do a LOT of stuff
  - We covered just a small fraction of what it can do
  - Really good for generating high-quality, customizable graphs with ease
  - Feel free to look up the documentation and learn more types of graphs etc.